

TYUMYATI, Russian Federation

WMO: 218130

Lat: 71.917N

Lon: 123.567E

Elev: 27

StdP: 101.01

Time Zone: 9.00 (E09)

Period: 90-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-51.1	-49.6	-53.2	0.0	-50.9	-51.6	0.0	-49.3	8.8	-25.9	8.1	-27.3	0.3	180	N/A

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	7.5	24.2	16.3	21.4	15.0	18.9	13.7	17.4	23.0	15.6	20.4	13.9	18.0	3.5	90

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
14.9	10.6	20.3	13.1	9.4	18.3	11.6	8.5	16.3	49.2	23.2	44.0	20.8	39.2	18.0	21.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 10.2	8.5	7.2	-52.2	28.2	2.3	1.9	-53.8	29.6	-55.2	30.7	-56.5	31.7	-58.1	33.0		
(5)			-52.2	18.6	2.3	0.9	-53.8	19.3	-55.2	19.8	-56.5	20.3	-58.1	20.9		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	-12.6	-36.0	-35.1	-24.3	-13.9	-4.5	8.0	10.6	9.8	2.8	-10.6	-26.2	-33.6
	DBStd	18.61	8.06	8.79	9.73	7.08	5.99	5.77	4.80	3.53	4.06	6.50	8.14	8.05
	HDD10.0	8417	1424	1262	1063	717	452	106	49	48	221	638	1087	1351
	HDD18.3	11307	1683	1495	1321	967	709	313	245	264	467	897	1337	1609
	CDD10.0	161	0	0	0	0	1	47	69	41	3	0	0	0
	CDD18.3	11	0	0	0	0	0	4	7	0	0	0	0	0
	CDH23.3	107	0	0	0	0	0	39	66	1	0	0	0	0
	CDH26.7	19	0	0	0	0	0	6	13	0	0	0	0	0
Wind	WSAvg	2.9	2.0	2.2	2.9	3.5	3.8	3.7	3.4	3.2	2.9	2.7	2.3	2.2
Precipitation	PrecAvg	375	25	17	20	16	29	36	47	52	41	39	25	28
	PrecMax	533	67	61	42	49	58	67	114	101	81	61	57	75
	PrecMin	268	8	2	5	1	12	8	17	22	17	12	8	7
	PrecStd	63	15	12	10	11	12	16	21	19	17	12	13	19
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	-14.2	-12.2	-1.2	3.5	14.0	28.1	28.9	23.2	19.8	1.1	-3.8	-9.8
		MCWB	-14.6	-12.8	-1.8	0.9	8.7	17.4	19.5	16.2	13.9	0.2	-4.4	-9.8
	2%	DB	-18.3	-16.2	-3.8	1.1	9.3	24.1	25.4	21.2	12.7	0.1	-7.7	-16.8
		MCWB	-18.8	-16.7	-4.6	-0.9	5.3	15.5	17.8	15.0	9.3	-0.5	-8.3	-17.2
	5%	DB	-21.0	-19.4	-6.3	-0.7	6.1	20.8	21.8	18.8	10.2	-1.3	-11.9	-18.9
		MCWB	-21.5	-19.7	-7.1	-2.4	3.3	13.9	15.8	14.0	8.0	-1.9	-12.4	-19.3
	10%	DB	-23.8	-22.8	-9.4	-2.8	3.8	18.0	18.8	16.3	8.2	-2.6	-14.7	-20.9
		MCWB	-24.1	-23.2	-10.1	-4.2	2.0	12.3	14.0	12.6	6.4	-3.2	-15.1	-21.3
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	-14.5	-12.7	-2.0	1.1	9.4	18.2	20.5	17.2	14.0	0.5	-4.2	-9.9
		MCDB	-14.2	-12.0	-1.4	3.0	13.6	26.2	27.7	21.4	19.9	0.8	-3.7	-9.8
	2%	WB	-18.7	-16.8	-4.6	-0.5	5.6	16.4	18.4	15.9	10.1	-0.5	-8.1	-17.1
		MCDB	-18.3	-16.2	-4.0	0.9	9.1	22.5	23.7	20.2	11.6	0.0	-7.7	-16.9
	5%	WB	-21.4	-19.7	-7.1	-2.4	3.8	14.4	16.4	14.5	8.1	-1.7	-12.4	-19.2
		MCDB	-21.0	-19.4	-6.3	-0.7	6.0	20.6	21.3	17.8	9.7	-1.2	-11.9	-18.9
	10%	WB	-24.1	-23.1	-10.0	-4.1	2.1	12.5	14.1	12.9	6.5	-3.2	-15.1	-21.3
		MCDB	-23.8	-22.9	-9.5	-2.7	3.7	17.8	17.7	16.1	8.2	-2.7	-14.7	-20.9
Mean Daily Temperature Range	5% DB	MDBR	7.6	7.7	10.3	11.4	6.8	8.1	7.5	7.6	5.6	5.7	7.3	7.2
		MCDBR	9.2	9.6	9.8	11.4	8.8	12.9	12.2	13.0	9.5	4.7	9.5	9.5
		MCWBR	9.0	9.4	9.7	10.1	5.4	6.6	6.2	7.6	6.3	4.7	9.4	9.3
	5% WB	MCDBR	9.3	9.8	10.1	11.2	7.8	12.6	11.1	11.6	8.7	5.0	9.7	9.5
		MCWBR	9.1	9.6	10.1	10.0	5.0	7.5	6.2	7.6	6.2	4.9	9.6	9.4
Clear-Sky Solar Irradiance	taub	N/A	0.197	0.277	0.308	0.339	0.343	0.391	0.375	0.297	0.205	N/A	N/A	
	taud	N/A	1.958	2.044	2.080	2.139	2.400	2.295	2.302	2.401	1.992	N/A	N/A	
	Ebn at Noon	0	441	672	794	820	839	769	725	689	397	0	0	
	Edh at Noon	0	32	77	107	117	95	102	89	56	29	0	0	
All-Sky Solar Radiation	RadAvg	0.00	0.22	1.66	4.17	5.66	5.44	4.80	3.08	1.41	0.35	0.00	0.00	
	RadStd	0.00	0.03	0.11	0.17	0.29	0.49	0.30	0.36	0.17	0.05	0.00	0.00	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (5 neighbors)	+1.05	N/A	N/A	N/A	N/A	N/A	-413	-394	N/A	N/A

Nomenclature: See separate page